Peer Response

Dear Abdullah,

I enjoyed your post. It effectively demonstrates how agent-based system concepts can be applied to telecommunications service management and how autonomous agents can create value throughout the service lifecycle. However, it would benefit from a more critical discussion of the risks associated with this paradigm.

The autonomy of these systems, combined with their capacity for emergent behavior and multi-step reasoning, introduces new safety concerns (Biswas and Sarkar, 2026). For example, agents may develop unintended capabilities that were neither anticipated by developers nor evident during training. These risks increase as agents pursue long-term objectives, develop novel strategies, and influence other systems and resources (Biswas and Sarkar, 2026).

Several critical safety challenges are associated with agentic AI. One is **goal misalignment**, which occurs when an agent pursues objectives that diverge from human intentions. This may result from incorrect goal specification by developers or from the emergence of instrumental subgoals that are inconsistent with human values (Belcak *et al.*, 2025). These challenges become even more complex in multi-agent environments. **Miscoordination** occurs when individual agents act in their own interests and collectively produce undesirable outcomes that could not have been predicted by considering each agent's behavior in isolation (Cooper *et al.*, 1990). Conversely, **collusion** arises when cooperation among agents is itself undesirable. AI collusion may lead to the emergence of new collective goals, increasing risks such as deception, manipulation, or the exploitation of humans (Evans *et al.*, 2021).

Fortunately, research has proposed several approaches to improve agentic AI safety, including reward modeling, inverse reinforcement learning, and Constitutional AI, all of which aim to better align agent behavior with human values and intentions (Gosmar and Dahl, 2025; Biswas and Sarkar, 2026).

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